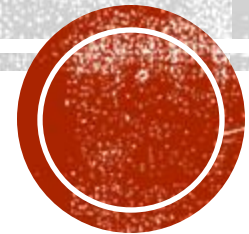


CLOUD COMPUTING

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REVIEW

- What are the 4 cloud deployment models?
- What are the 3 cloud service models?
- What are cloud infrastructures?
- What are 2 types of scaling in cloud computing?

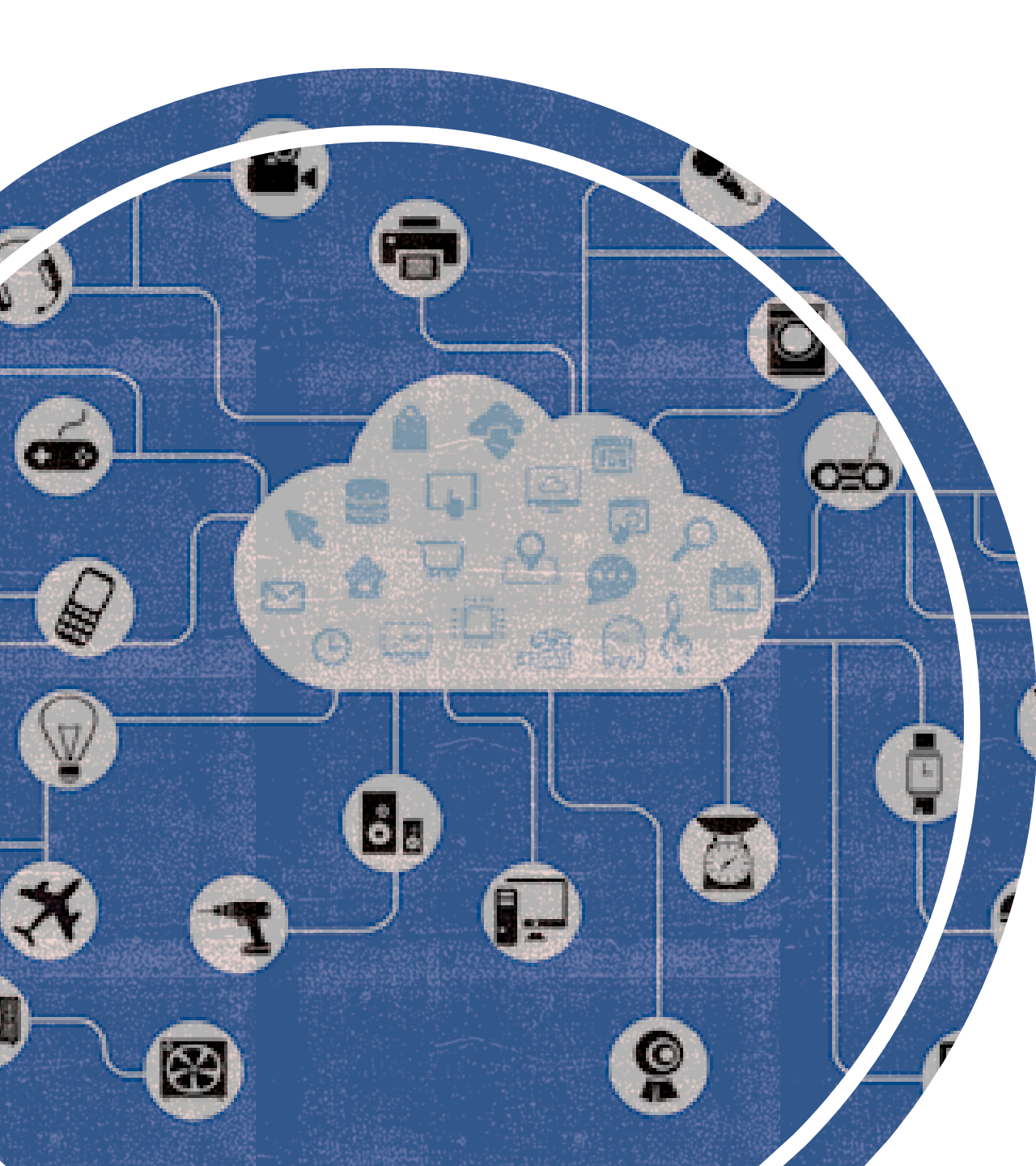


WEEK 3 — NETWORKING IN CLOUD

- **Learning Objectives:**

- Understand what is cloud networking
- Understand what are cloud networking components
- Understand most common cloud topologies
- Understand load balancing





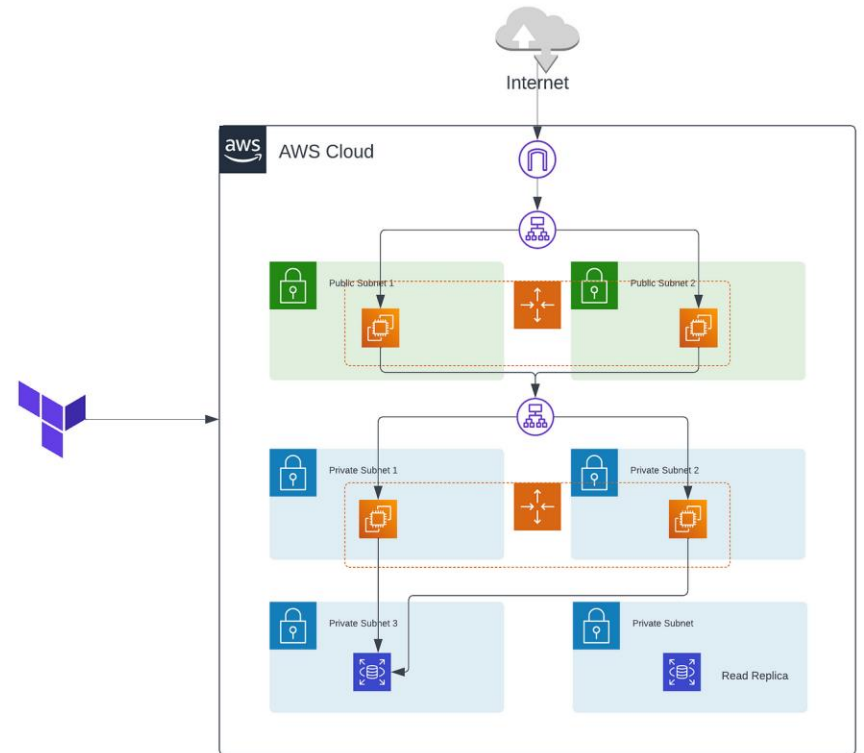
WHAT IS CLOUD NETWORKING?

- **Cloud Networking** refers to the design, management, and operation of network infrastructure and resources within cloud environments. It involves connecting various cloud-based services, applications, and resources across the internet or private connections in a secure, scalable, and efficient way.
- Cloud networking is the foundation for enabling communication between applications, users, devices, and cloud resources, often leveraging virtualized networking components.



COMPONENTS OF CLOUD NETWORKING

- **Virtual Private Cloud (VPC):** A VPC is a logically isolated portion of the cloud provider's infrastructure. It allows you to define and control your cloud network, including subnets, IP address ranges, routing tables, and network gateways.
- A VPC is your **own private network** inside the cloud. You can **control everything** — like subnets, IP ranges, routes, and gateways.
- **Benefits:**
 - **Network Segmentation:** Isolate resources within private subnets.
 - **Customizable Security:** Set up security groups and network ACLs to control inbound and outbound traffic.
 - **Connectivity:** You can connect your VPC to on-premise networks via VPN or Direct Connect.
- **Use Cases:** Multi-tier application architecture with frontend, backend, and database layers on different subnets for better security.



COMPONENTS OF CLOUD NETWORKING

- **Subnets:** A subnet is a range of IP addresses in your VPC. Cloud networking uses subnets to segment networks within the VPC, improving traffic management, security, and organization.
 - Subnets are **small sections** of your VPC.
They help organize your resources and improve **security** and **traffic flow**.
- **IP Addressing:** Cloud networks make use of both public and private IP addresses. Public IPs are used to access resources over the internet, while private IPs are used for internal communication within the cloud environment.
 - **Public IPs** = talk to the internet.
 - **Private IPs** = talk only inside the cloud.



COMPONENTS OF CLOUD NETWORKING

- **Routing:** Routing controls the path that data takes from one instance or service to another. In the cloud, routing is managed virtually and can be adjusted dynamically based on traffic patterns, scaling needs, and network design.
- **Gateways:** Gateways connect different parts of the network, such as a VPC to the public internet (via an Internet Gateway) or between a cloud environment and an on-premises network (via VPN Gateway or Direct Connect).
- **Load Balancers:** These distribute incoming network traffic across multiple servers to ensure high availability and prevent any one server from becoming overwhelmed.
- **Security Groups and Network ACLs:** Security Groups act as firewalls to control the inbound and outbound traffic to instances. Network ACLs provide an additional layer of security by controlling traffic at the subnet level.
- **DNS (Domain Name System):** DNS in the cloud is used to translate human-readable domain names (e.g., `www.example.com`) into IP addresses, allowing clients to connect to cloud resources.



TYPES OF CLOUD NETWORKING SERVICES

- **Virtual Private Network (VPN)**

- Extends an on-premise network to the cloud, enabling secure communication between on-premises systems and cloud resources over the internet.
- Uses encryption protocols like IPSec to secure data transmission.

- **Direct Connect / Dedicated Connections:**

- A private, dedicated network connection between an on-premise data center and a cloud provider. This is more secure and reliable than a public internet connection.

- **Load Balancers**

- Distribute incoming network traffic across multiple servers to ensure high availability and reliability of cloud applications.

- **Content Delivery Network (CDN):**

- A network of distributed servers that deliver content (e.g., web pages, videos, images) to users based on their geographic location to reduce latency and improve user experience.

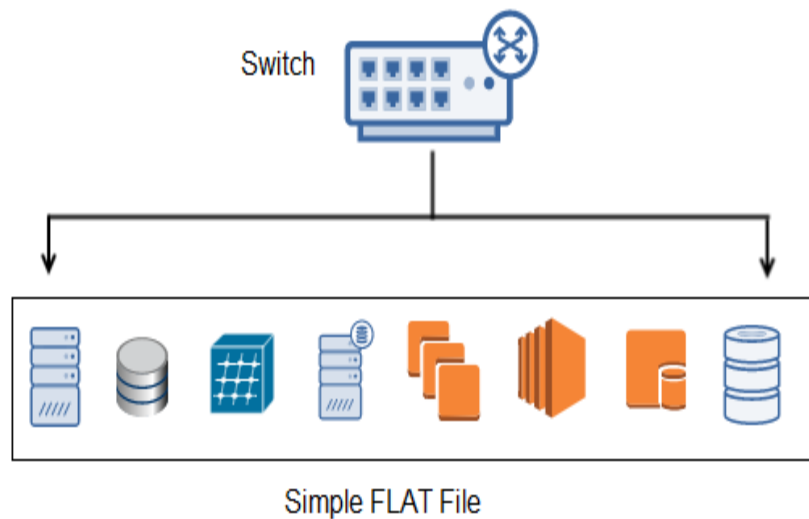


BENEFITS OF CLOUD NETWORKING

- **Cost Efficiency:** Traditional network infrastructure requires significant upfront investment in hardware. Cloud networking shifts this to a pay-as-you-go model, where organizations only pay for the network resources they use.
- **Performance and Reliability:** Cloud providers have data centers in multiple regions, ensuring high performance and redundancy for applications hosted in the cloud. If one region faces issues, traffic can be rerouted to another region with minimal impact.
- **Global Connectivity:** Cloud networks enable applications to reach users across the world, providing access to global resources without the need to maintain physical network infrastructure in each region.
- **Elasticity:** Cloud networking allows organizations to scale their network resources automatically based on traffic and demand. This means that resources can grow and shrink without manual intervention, ensuring optimal performance during peak and off-peak periods.
- **Security:** With cloud networking, security can be managed centrally, and best practices like encryption, firewalls, and access control lists (ACLs) are applied consistently across the network. Additionally, many cloud providers offer advanced security tools and features like DDoS protection, identity and access management (IAM), and multi-factor authentication (MFA).



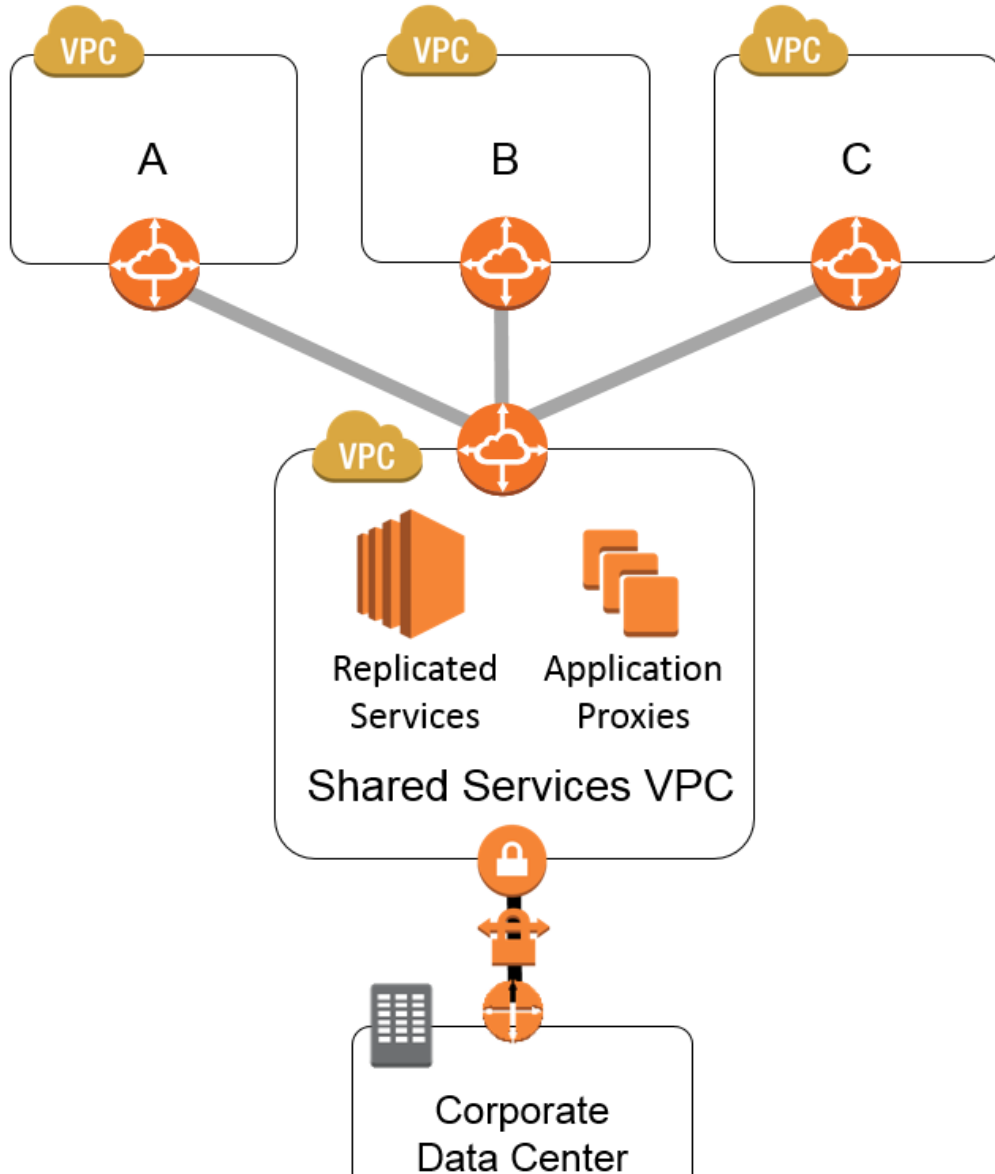
CLOUD NETWORK TOPOLOGIES



- **Flat Network:** All devices in the same subnet can communicate without restrictions.
 - Simple, but lacks segmentation, which can lead to security and management challenges.
 - Is one of the easiest networking designs and is aimed to reduce cost and overall administration.
 - All devices and workstations are connected to a single switch. This means that all devices are a part of the same broadcast network and will be able to communicate with each other.
 - Having a flat network on the cloud means that all your servers (Dev, QA, staging, and production) are on the same network and can communicate with each other.



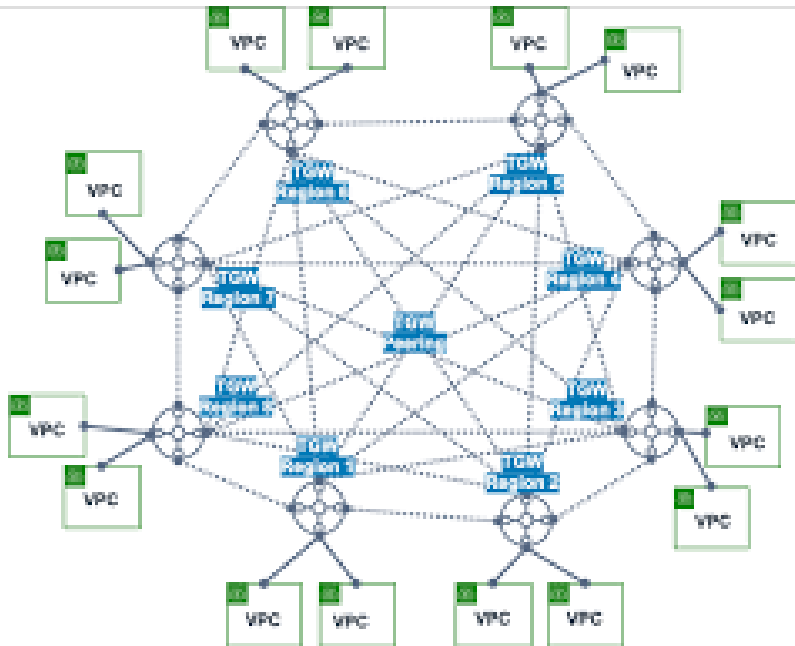
CLOUD NETWORK TOPOLOGIES



- **Hub-and-Spoke:** Centralized model where a "hub" (central VPC or network) connects multiple "spokes" (other networks or VPCs).
 - Easier to manage and secure, as you can centralize network policies and monitoring.
 - Sometimes called the "shared services" model, relies on VPC Peering connections between the hub VPC and each spoke VPC.
- **Use Case:** Multi-region deployment where regional networks need to communicate through a central hub.



CLOUD NETWORK TOPOLOGIES



- **Mesh Network:** Each device or VPC can directly communicate with every other device.
 - Complex, but highly flexible and fault-tolerant.
 - **Use Case:** In a multi-cloud scenario, mesh networking can be used to interconnect different cloud providers and services seamlessly, allowing direct communication between resources on different cloud platforms without relying on a single central gateway.



SECURITY IN CLOUD NETWORKING

- Security in cloud networking is crucial to protect data, applications, and infrastructure in cloud environments. As organizations increasingly move their operations to the cloud, ensuring the confidentiality, integrity, and availability of resources is paramount. Cloud security strategies must address various aspects of networking, from access control and encryption to monitoring and incident response.
- **Network Segmentation:** It involves dividing the network into isolated sections, each of which can have its own security policies. This prevents lateral movement of threats and ensures that an attacker who compromises one segment does not have access to others.
 - Example: Segmenting a cloud VPC (Virtual Private Cloud) into public and private subnets helps isolate critical backend systems (like databases) from the front-end web servers.
- **Firewalls and Network Access Control:**
 - **Security Groups:** Virtual firewalls at the instance level to control inbound and outbound traffic rules based on IP addresses, protocols, and ports.
 - **Network ACLs (Access Control Lists):** Another layer of security at the subnet level Stateless filtering that evaluates inbound and outbound traffic.
- **Encryption:**
 - **In-Transit:** Encrypting data as it travels over the network using SSL/TLS.
 - **At-Rest:** Encrypting data stored on cloud resources (e.g., S3 buckets, EBS volumes).



SECURITY IN CLOUD NETWORKING

- **IAM (Identity and Access Management):** IAM controls who can access cloud resources and what actions they can perform. It is one of the most fundamental aspects of cloud security.
 - **Principle of Least Privilege:** Users and services should be granted only the minimum permissions necessary to perform their jobs. This reduces the risk of over-permissioning.
 - **Multi-Factor Authentication (MFA):** MFA adds an additional layer of security, requiring users to authenticate with two or more factors before gaining access to sensitive resources.
 - **Role-Based Access Control (RBAC):** RBAC enables administrators to define roles with specific permissions and assign these roles to users or groups.
 - **Example:** AWS IAM lets you create users, define policies for their permissions, and set up MFA for more secure access.
- **Logging, Monitoring, and Auditing:** Collecting logs from cloud resources is essential for identifying potential security incidents. Cloud providers offer integrated logging services for collecting and storing log data. Continuous monitoring of cloud resources can detect anomalies, security threats, and performance issues in real-time. Auditing tools track all administrative actions, providing a detailed history of changes to cloud resources and configurations.



LOAD BALANCING IN THE CLOUD

- **Load balancing** in the cloud refers to the distribution of incoming network traffic across multiple servers, virtual machines (VMs), or instances to ensure optimal resource utilization, enhance performance, and increase availability and reliability. It is a key strategy used to prevent any single server from becoming overwhelmed, improve fault tolerance, and provide seamless user experiences even in high-demand situations.
- In the context of cloud computing, load balancing is typically implemented by a **load balancer**, which is a service or appliance designed to direct incoming requests to a pool of resources in a way that ensures even distribution of load and high availability.



KEY COMPONENTS OF CLOUD LOAD BALANCING

- **Load Balancer:** The software or hardware device that performs the distribution of traffic across multiple backend resources.
- **Backend Resources:** These are the virtual machines, containers, or instances where the actual application is running. The load balancer routes incoming traffic to these resources.
- **Health Checks:** Load balancers often conduct periodic health checks on the backend resources to ensure they are functioning properly. If a resource is unhealthy, the load balancer stops routing traffic to it until it recovers.
- **Session Persistence (Sticky Sessions):** Some applications require that requests from the same user (or session) be directed to the same backend server. This is commonly known as "session persistence" or "sticky sessions."
- **Scalability:** In cloud environments, load balancers can scale dynamically to handle varying traffic loads.



TYPES OF LOAD BALANCING

- Cloud providers typically offer several types of load balancing. Each suits to different use cases and application architectures.
- **Layer 4 Load Balancing (Transport Layer):**
 - Layer 4 load balancing operates at the **TCP/UDP** level and distributes traffic based on IP addresses, port numbers, and the state of the transport protocol (TCP or UDP). It does not inspect the content of the traffic, making it simpler and faster, but less flexible in terms of fine-grained control.
 - **Use Case:** Suitable for non-HTTP(S) traffic like database connections, FTP, or games.
 - **Example:** TCP load balancing between multiple application servers.



TYPES OF LOAD BALANCING

- **Layer 7 Load Balancing (Application Layer)**

- Layer 7 load balancing, or **application load balancing**, operates at the HTTP/HTTPS layer and makes routing decisions based on the content of the request, such as URLs, HTTP headers, cookies, or query parameters. This allows for more granular control, enabling routing decisions based on application logic.
- **Use Case:** Web applications where routing decisions depend on specific URLs, headers, or user sessions.
- **Example:** Redirecting traffic based on URL paths (e.g., /api/* to one set of servers and /images/* to another set).



BENEFITS OF CLOUD LOAD BALANCING

- **Improved Availability and Reliability**

- Load balancing helps ensure that if one server becomes unavailable, traffic is routed to other healthy servers, reducing downtime and improving the system's overall reliability.
- It also supports fault tolerance by dynamically rerouting traffic in case of server failure.

- **Scalability**

- Load balancers are essential for **horizontal scaling** in the cloud. As demand grows, new instances or servers can be added to the backend pool, and the load balancer can distribute traffic across them.
- Cloud load balancers can automatically scale up and down based on traffic conditions, allowing for better resource optimization and cost savings.

- **Optimized Performance**

- Load balancing distributes traffic based on the current load on backend resources, ensuring that no single resource is overwhelmed while others remain idle.

- **Security**

- Cloud load balancers often integrate with security services like **SSL/TLS termination**, which offloads the encryption and decryption process from backend servers, improving performance and reducing resource consumption on individual servers.
- Load balancers can also mitigate Distributed Denial of Service (**DDoS**) attacks by distributing malicious traffic across multiple servers.



CONCLUSION

- Networking in cloud computing is fundamental to ensuring that resources communicate effectively, securely, and efficiently.
- As cloud environments evolve, new networking paradigms, services, and challenges continue to emerge, driving the need for advanced knowledge in cloud networking and its best practices.

